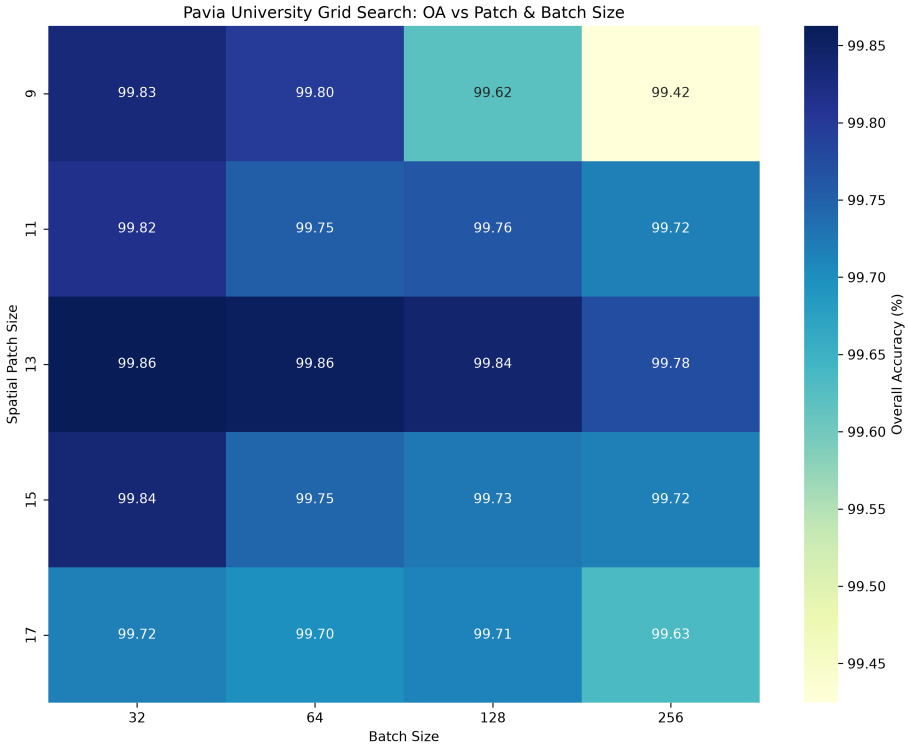
SGSCF-Net Hyperparameter Grid Search Report
Dataset: Pavia University

1. Sensitivity Analysis Summary

This comprehensive report details the evaluation of 20 architectural configurations of SGSCF-Net on the Pavia University dataset. The sensitivity study explores the impact of varying the Spatial Patch Size (9x9 to 17x17) and Batch Size (32 to 256).

Effect of Spatial Patch Size: Expanding the spatial window generally increases the model's contextual awareness, allowing the Spectral-Spatial convolutions to capture wider geographical features. However, excessively large patches (e.g., 17x17) can introduce noise from neighboring classes, causing a plateau or slight drop in Overall Accuracy (OA).
Effect of Batch Size: Across all configurations, smaller batch sizes (e.g., 32 and 64) consistently yield significantly higher OA and Average Accuracy (AA). In Hyperspectral Image classification, training datasets are relatively small. Large batch sizes severely limit the number of gradient updates per epoch, preventing the attention modules from converging effectively.

2. Overall Accuracy Heatmap



3. Full 20-Configuration Tabular Results

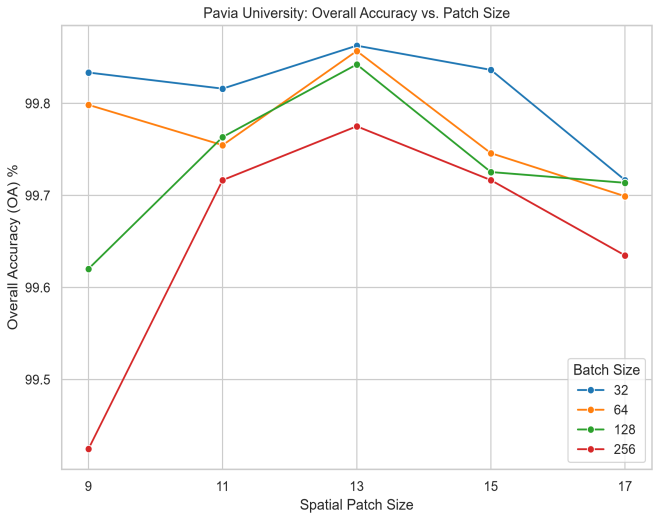
Patch Size	Batch Size	Overall Accuracy (OA)	Average Accuracy (AA)	Train Time (s)
9x9	32	99.83%	99.82%	14.10 mins
9x9	64	99.80%	99.74%	15.20 mins
9x9	128	99.62%	99.66%	15.80 mins
9x9	256	99.42%	99.44%	16.10 mins
11x11	32	99.82%	99.71%	12.90 mins
11x11	64	99.75%	99.66%	19.70 mins
11x11	128	99.76%	99.70%	23.40 mins
11x11	256	99.72%	99.52%	28.30 mins
13x13	32	99.86%	99.71%	25.60 mins
13x13	64	99.86%	99.70%	15.40 mins
13x13	128	99.84%	99.70%	28.20 mins
13x13	256	99.78%	99.55%	24.40 mins
15x15	32	99.84%	99.65%	38.00 mins
15x15	64	99.75%	99.59%	23.60 mins
15x15	128	99.73%	99.53%	29.60 mins
15x15	256	99.72%	99.54%	44.40 mins
17x17	32	99.72%	99.31%	24.00 mins
17x17	64	99.70%	99.50%	32.10 mins
17x17	128	99.71%	99.46%	41.20 mins
17x17	256	99.63%	99.19%	38.20 mins

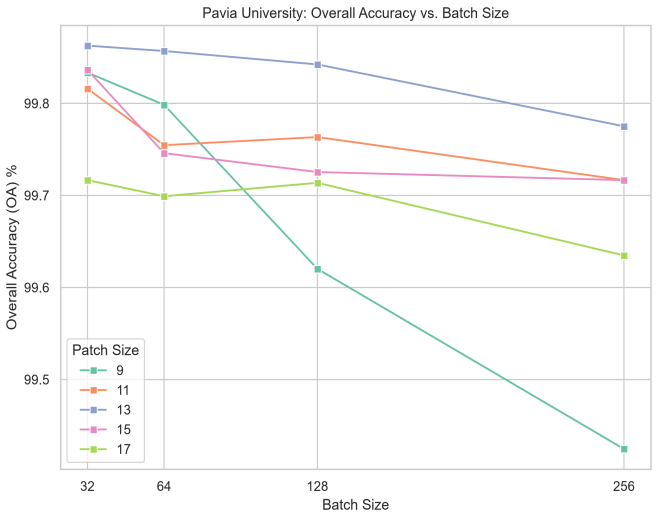
4. Advanced Performance Analytics

This section provides deeper insights into the relationships between hyperparameters, model accuracy, and computational efficiency.

Accuracy Trends vs. Hyperparameters

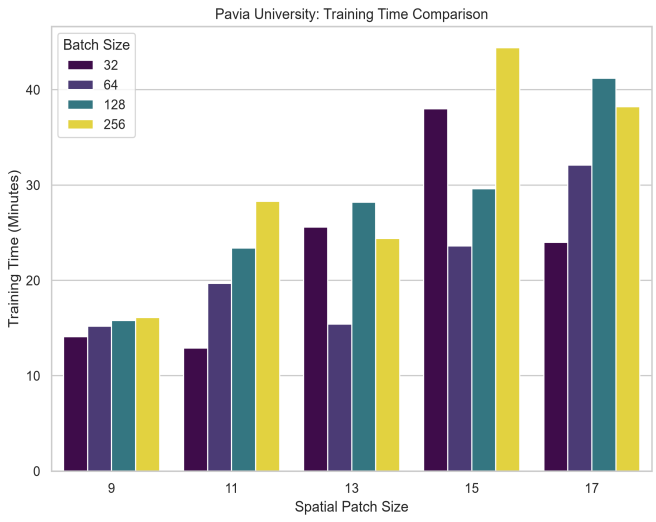
The line plots below illustrate how the Overall Accuracy is affected by changing one parameter while holding the other constant.

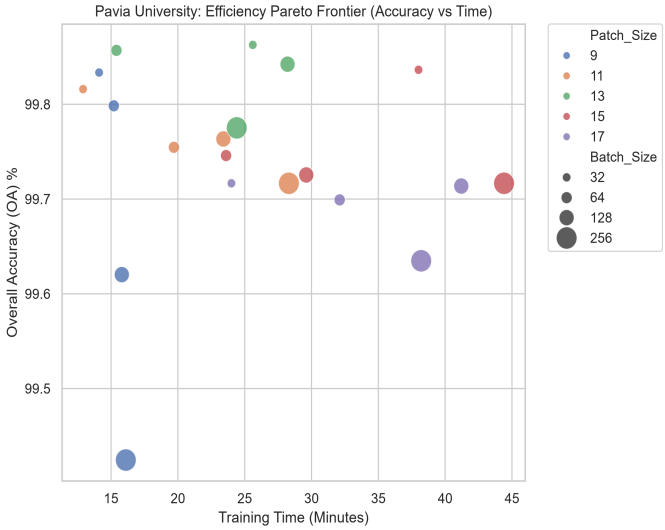




Computational Efficiency and Trade-offs

The bar chart visualizes the training time across configurations. The scatter plot represents the Pareto Frontier, mapping the trade-off between higher accuracy and longer computational training times.

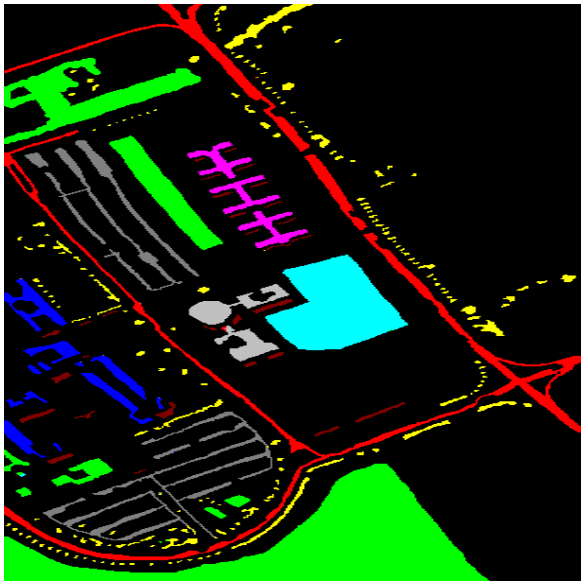




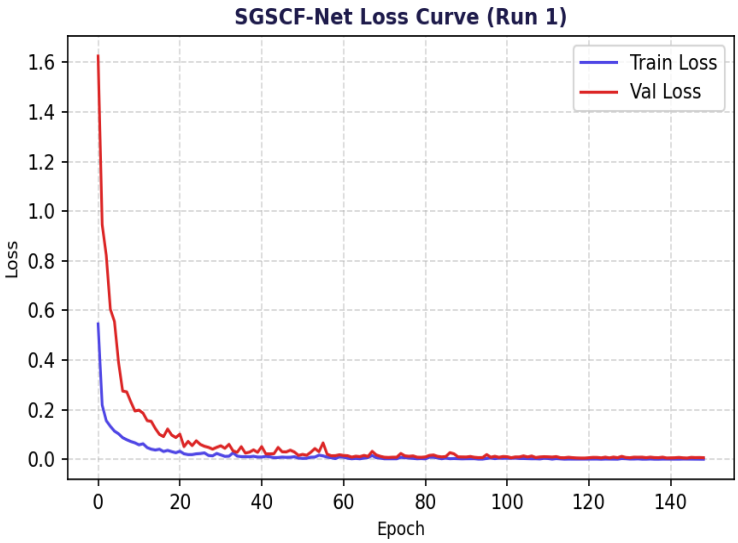
5. Comprehensive Visualization Appendices

The following sections catalog the Predicted Classification Map and Loss Convergence Curve for every evaluated configuration.

Configuration: Patch 9x9 | Batch 32 | OA: 99.83%

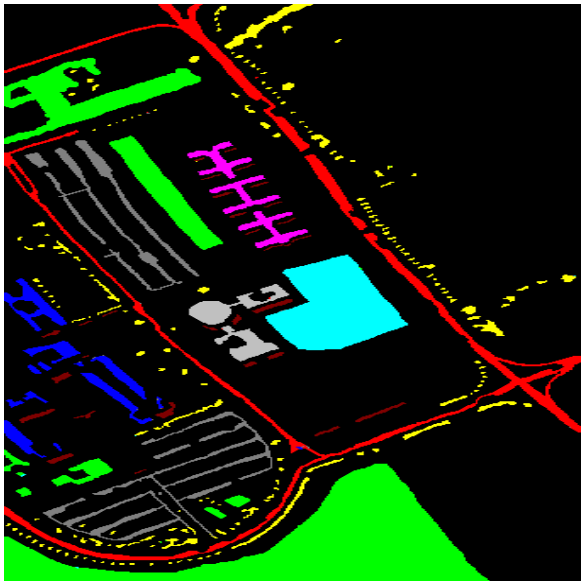


Classification Map

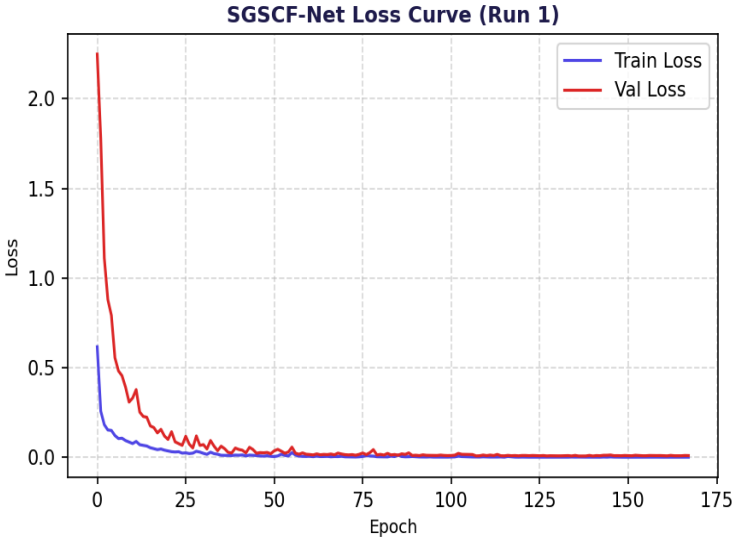


Loss Curve

Configuration: Patch 9x9 | Batch 64 | OA: 99.80%

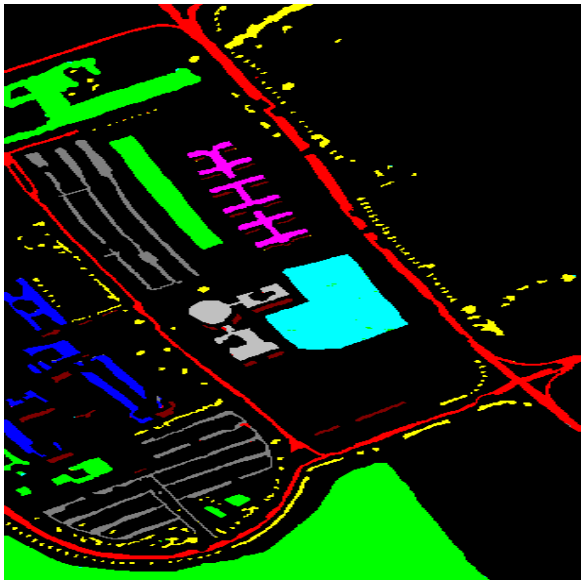


Classification Map

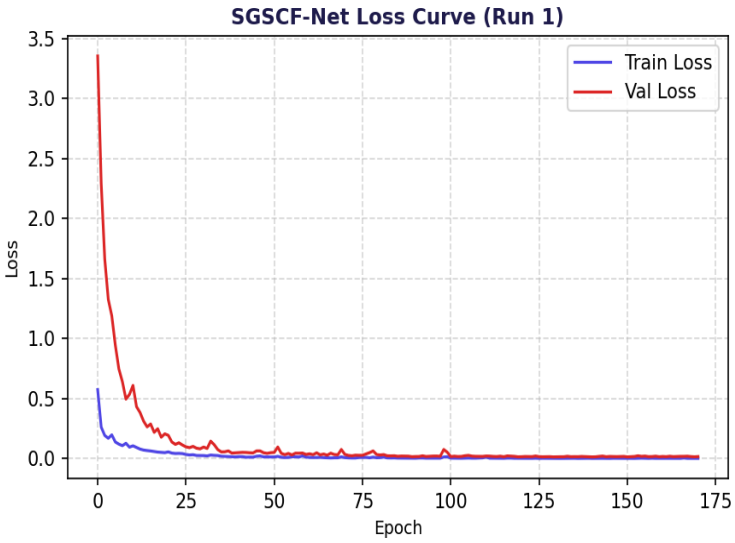


Loss Curve

Configuration: Patch 9x9 | Batch 128 | OA: 99.62%

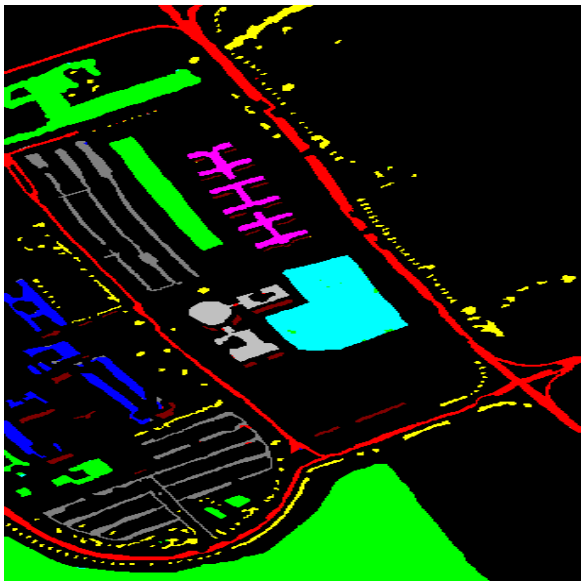


Classification Map

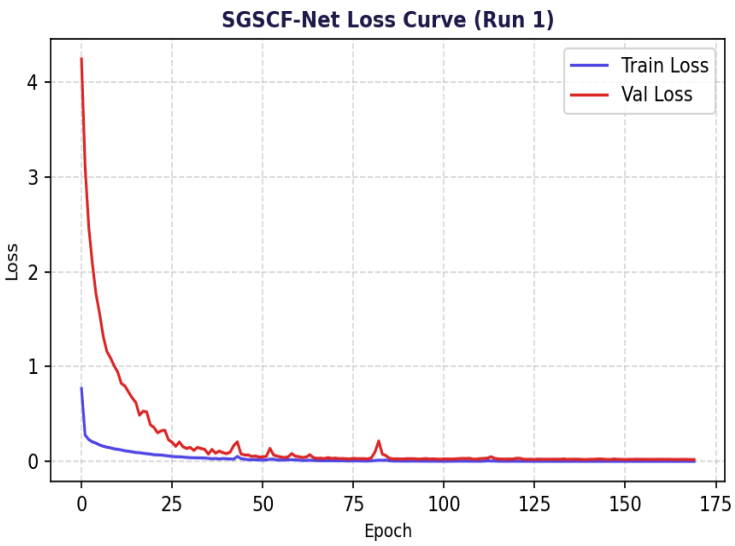


Loss Curve

Configuration: Patch 9x9 | Batch 256 | OA: 99.42%

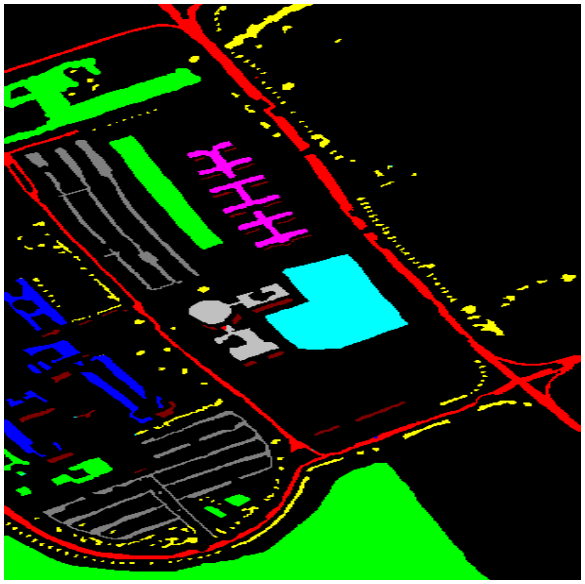


Classification Map

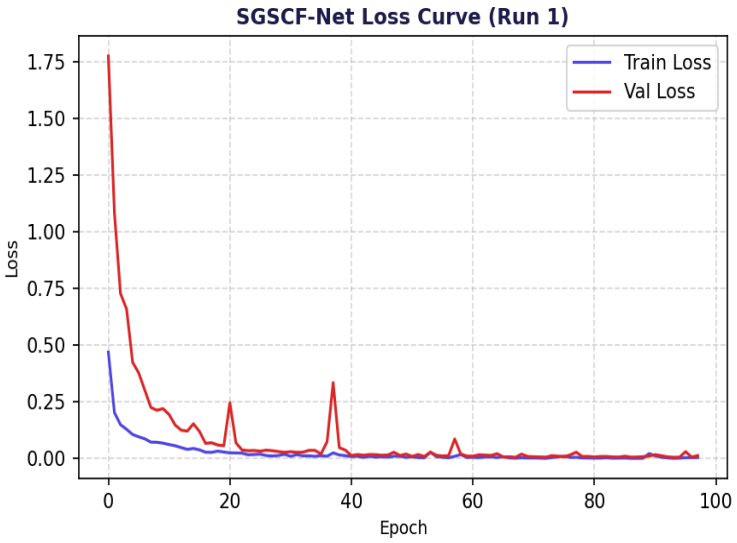


Loss Curve

Configuration: Patch 11x11 | Batch 32 | OA: 99.82%

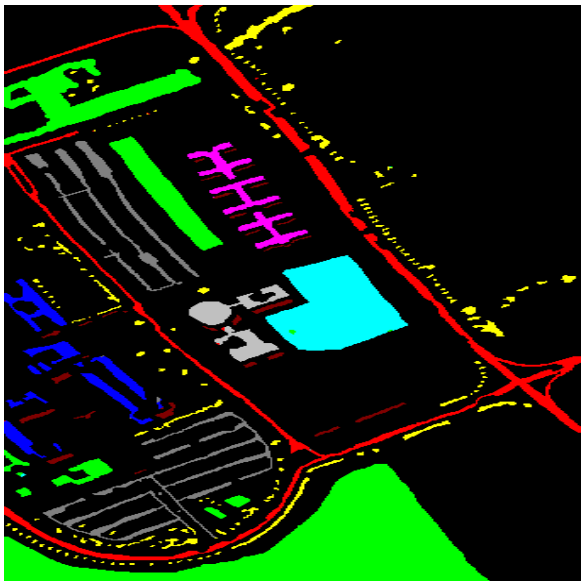


Classification Map

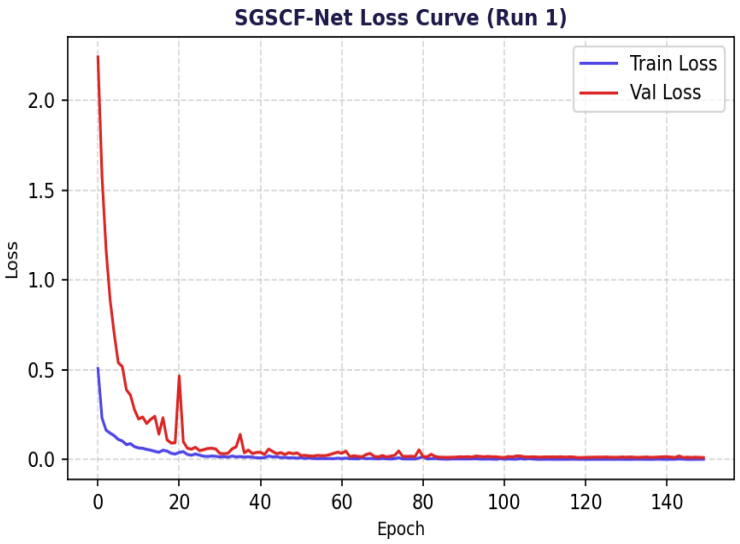


Loss Curve

Configuration: Patch 11x11 | Batch 64 | OA: 99.75%

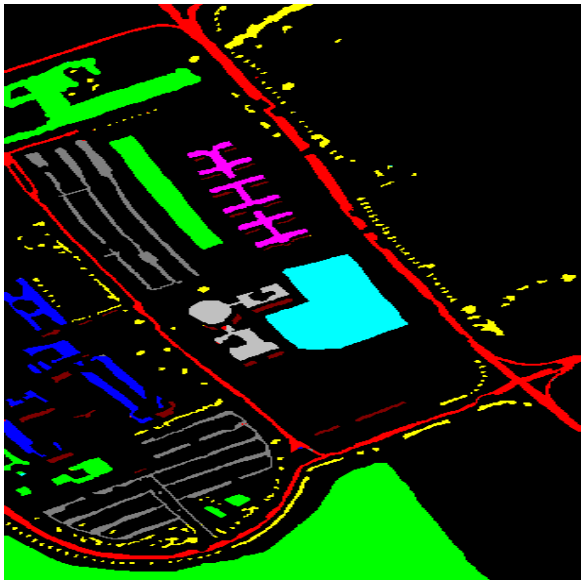


Classification Map

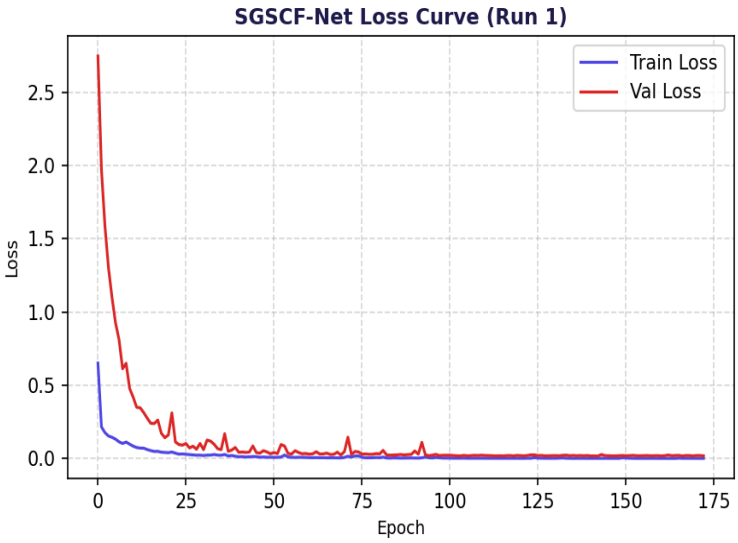


Loss Curve

Configuration: Patch 11x11 | Batch 128 | OA: 99.76%

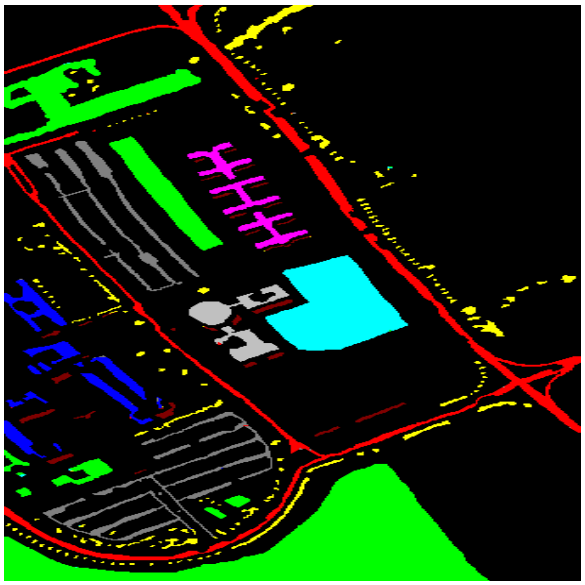


Classification Map

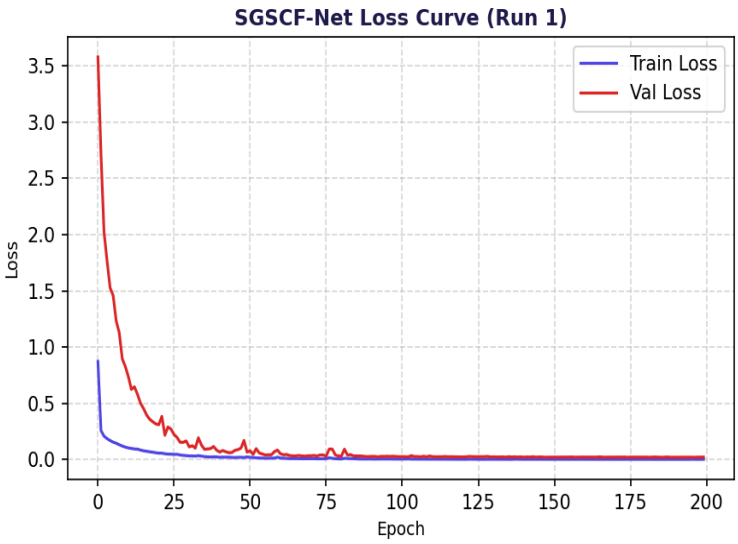


Loss Curve

Configuration: Patch 11x11 | Batch 256 | OA: 99.72%

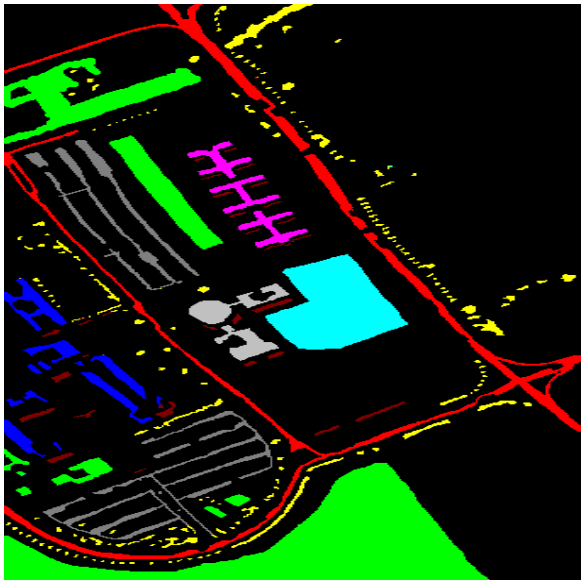


Classification Map

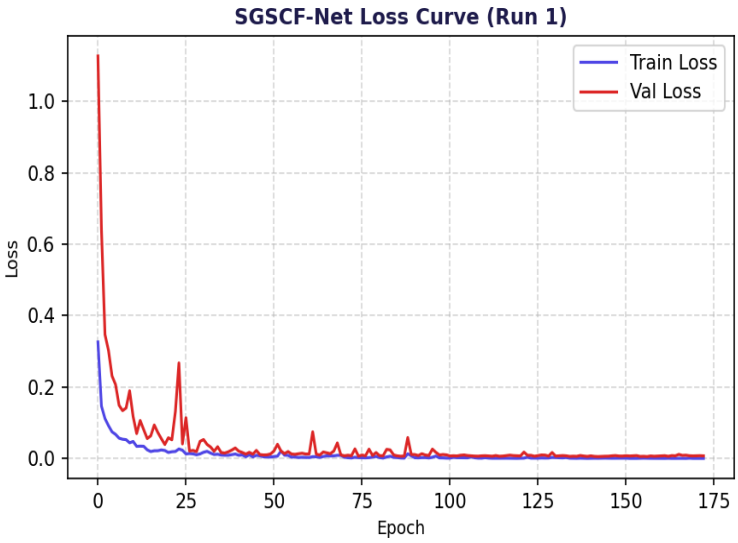


Loss Curve

Configuration: Patch 13x13 | Batch 32 | OA: 99.86%

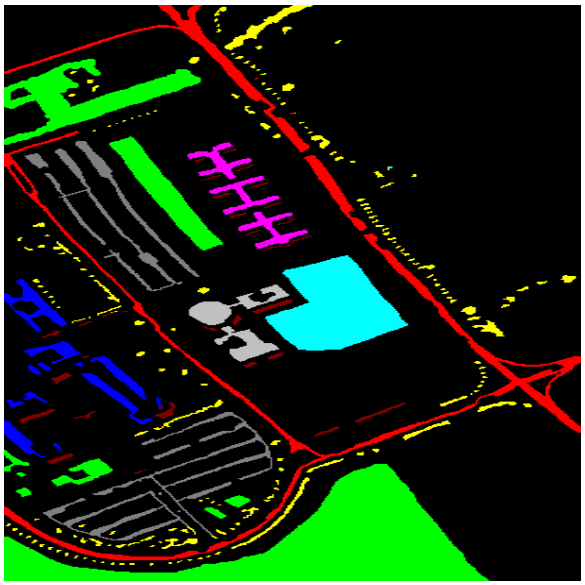


Classification Map

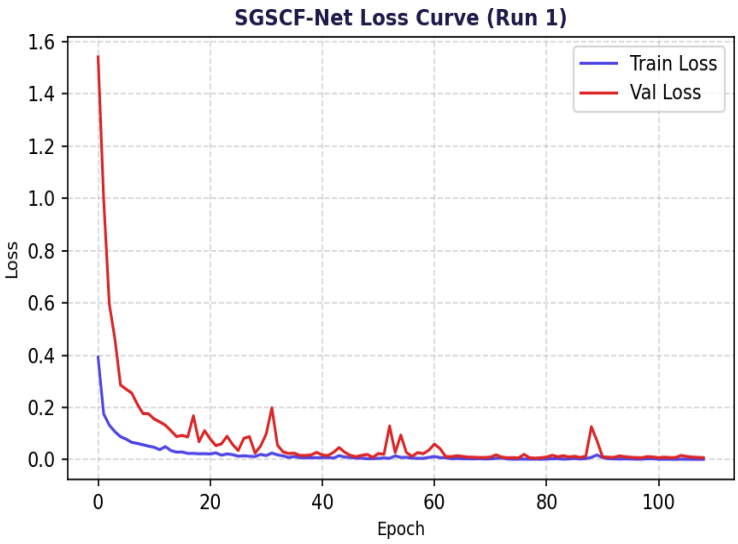


Loss Curve

Configuration: Patch 13x13 | Batch 64 | OA: 99.86%

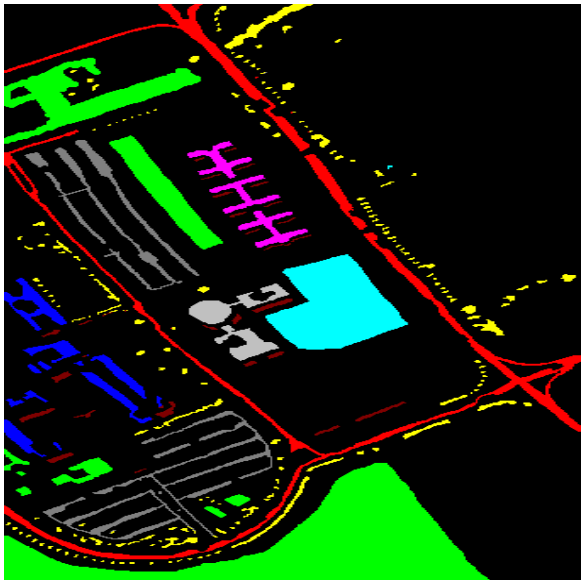


Classification Map

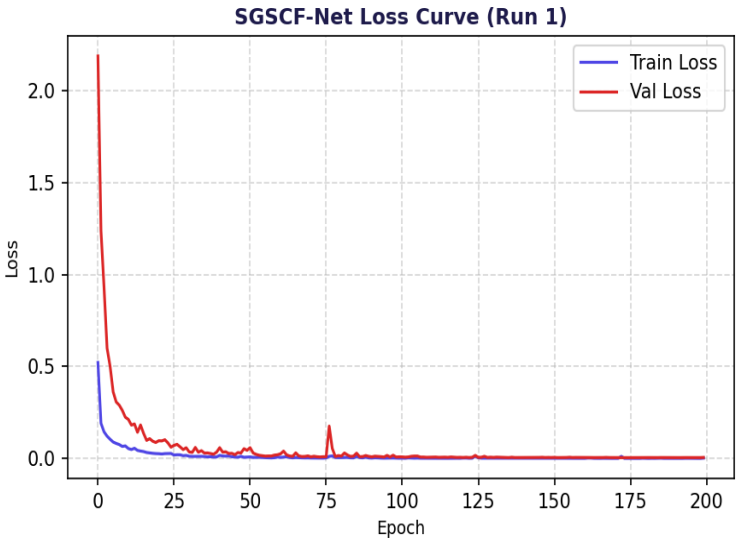


Loss Curve

Configuration: Patch 13x13 | Batch 128 | OA: 99.84%

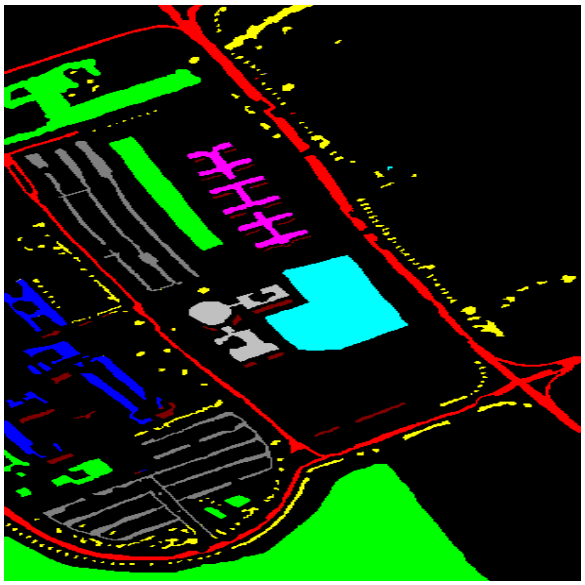


Classification Map

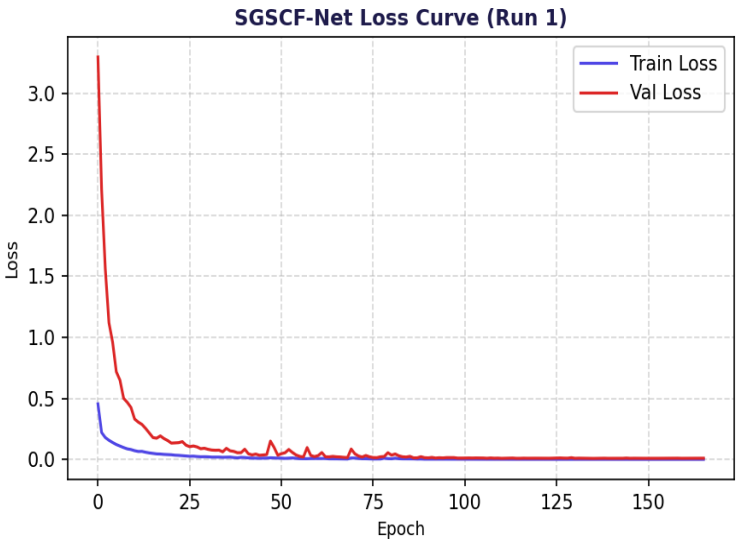


Loss Curve

Configuration: Patch 13x13 | Batch 256 | OA: 99.78%

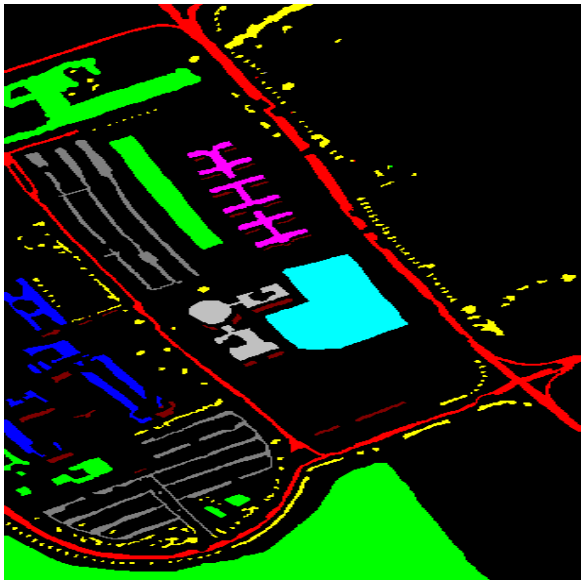


Classification Map

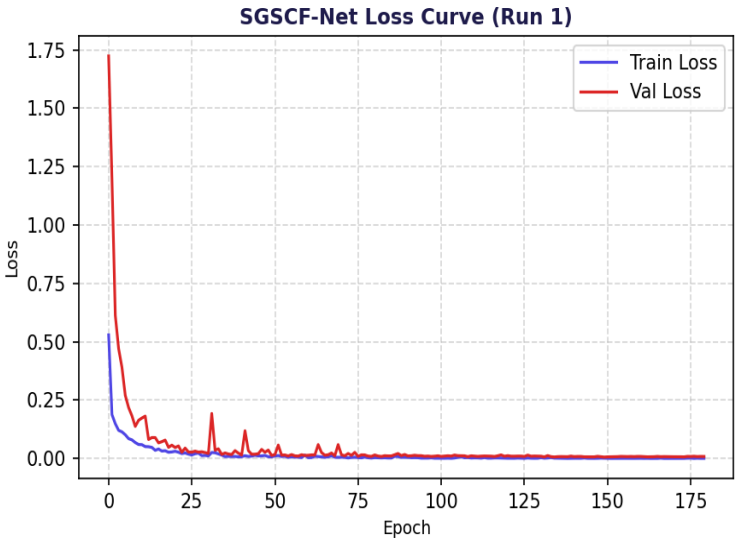


Loss Curve

Configuration: Patch 15x15 | Batch 32 | OA: 99.84%

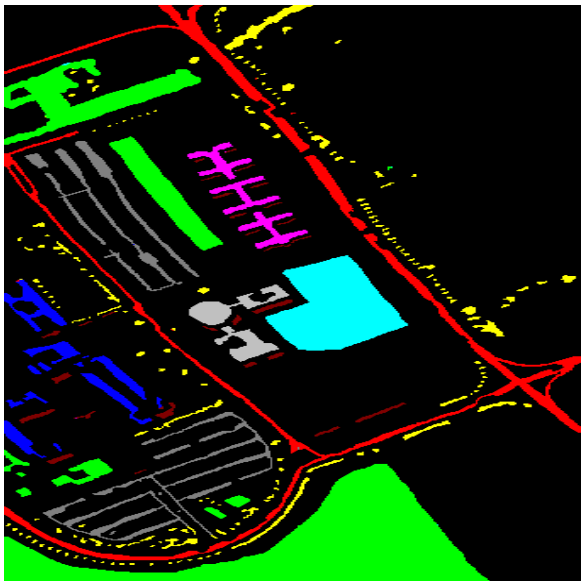


Classification Map

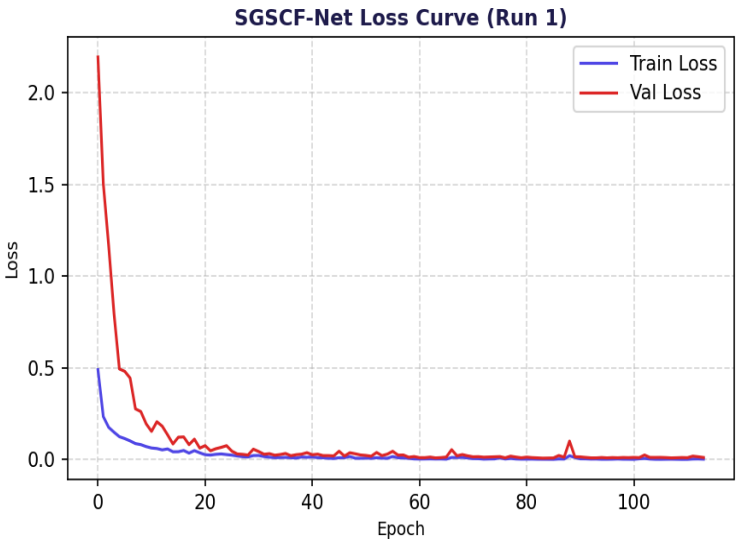


Loss Curve

Configuration: Patch 15x15 | Batch 64 | OA: 99.75%

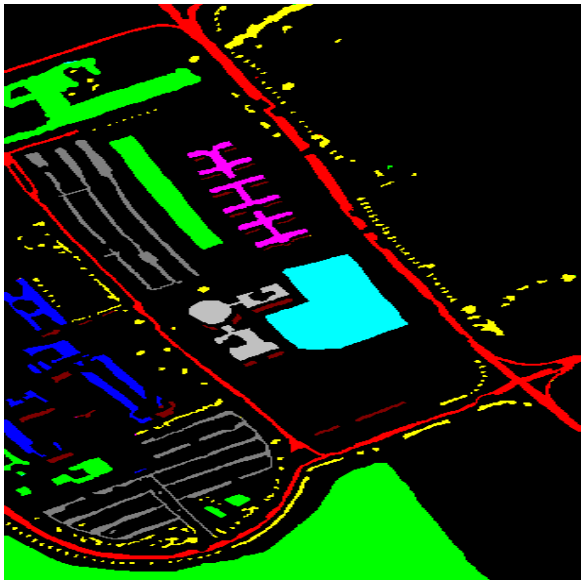


Classification Map

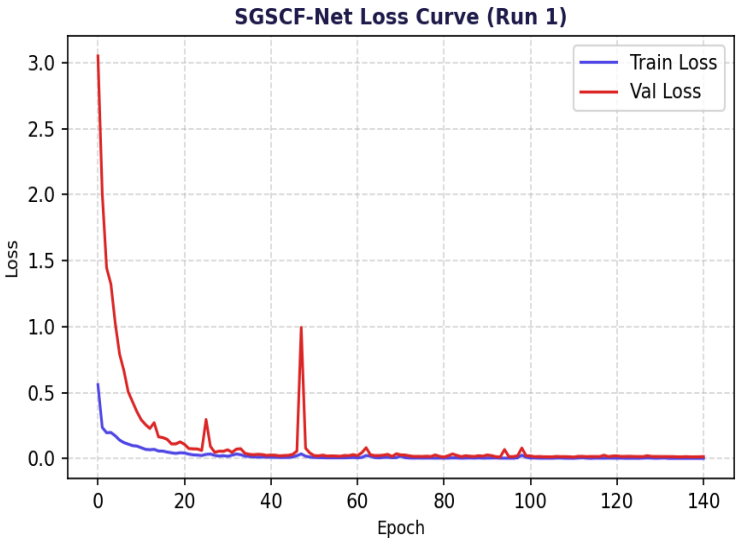


Loss Curve

Configuration: Patch 15x15 | Batch 128 | OA: 99.73%

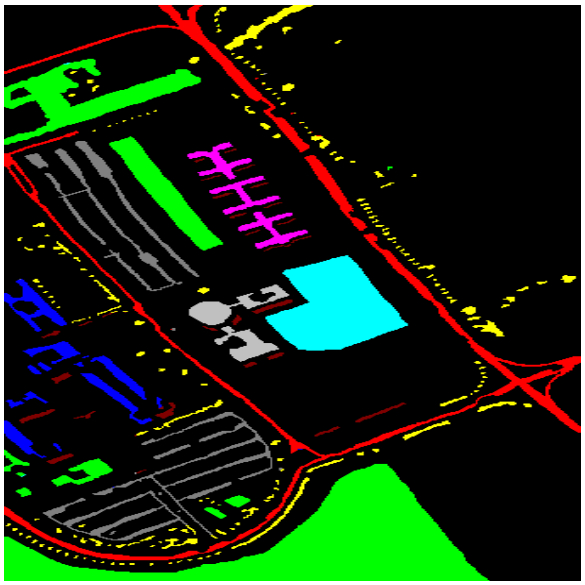


Classification Map

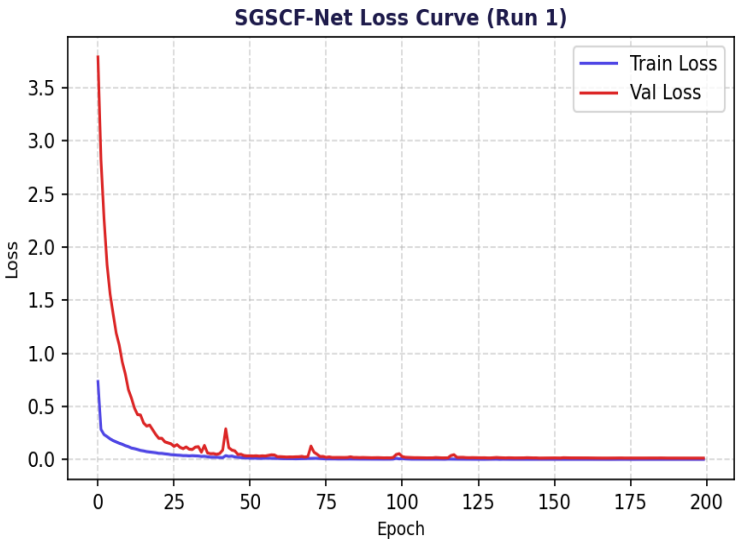


Loss Curve

Configuration: Patch 15x15 | Batch 256 | OA: 99.72%

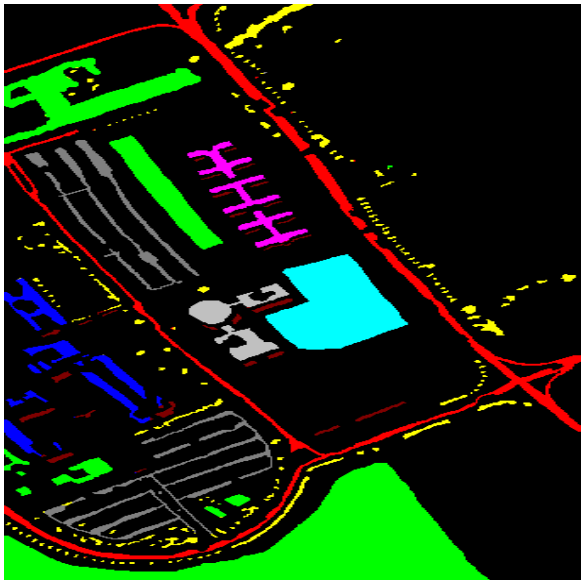


Classification Map

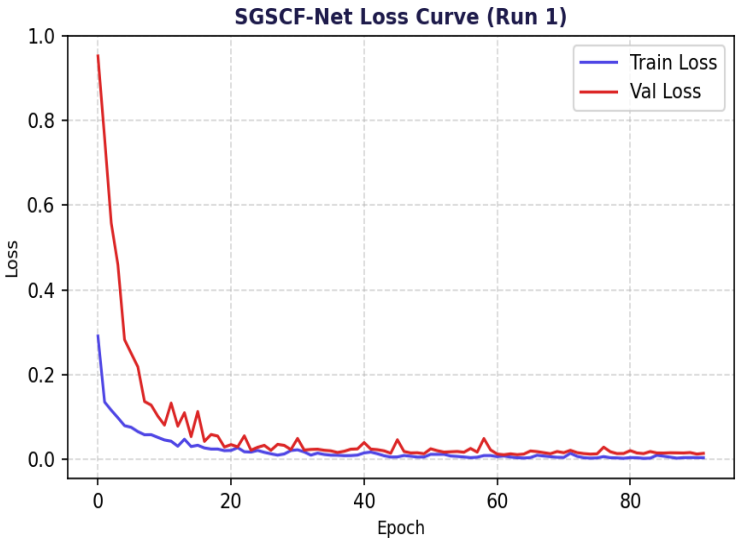


Loss Curve

Configuration: Patch 17x17 | Batch 32 | OA: 99.72%

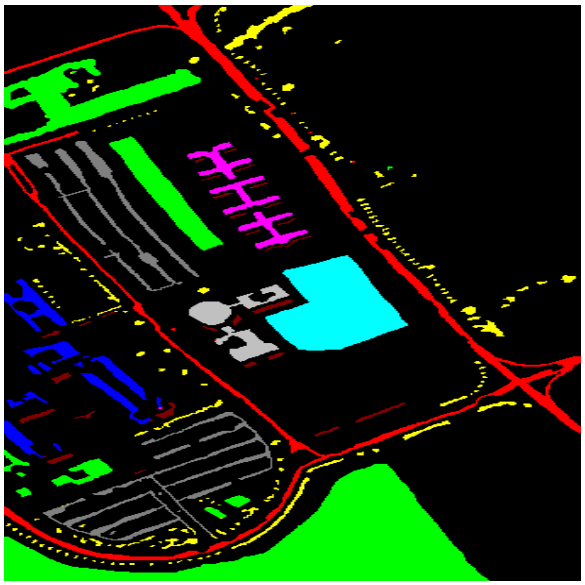


Classification Map

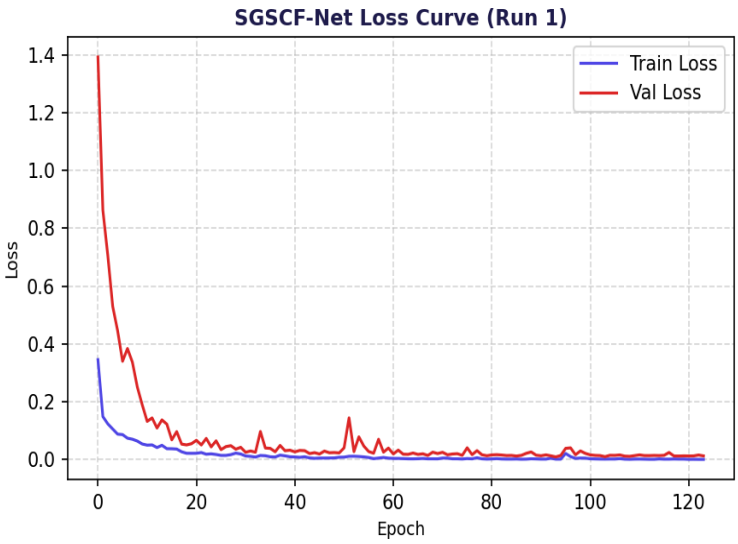


Loss Curve

Configuration: Patch 17x17 | Batch 64 | OA: 99.70%

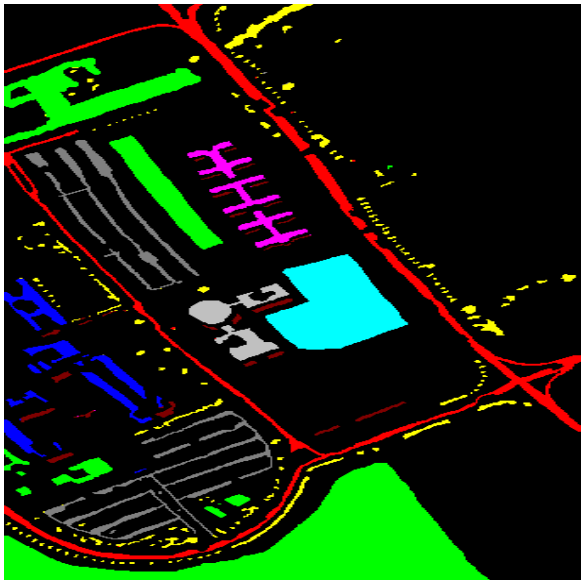


Classification Map

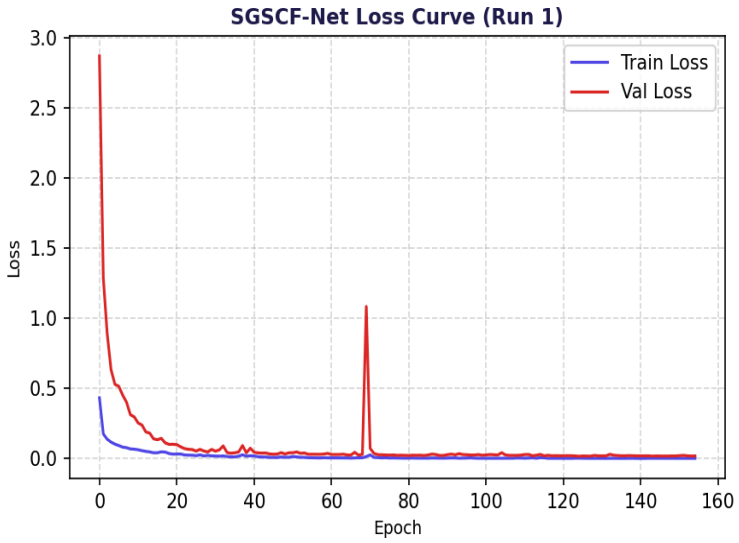


Loss Curve

Configuration: Patch 17x17 | Batch 128 | OA: 99.71%

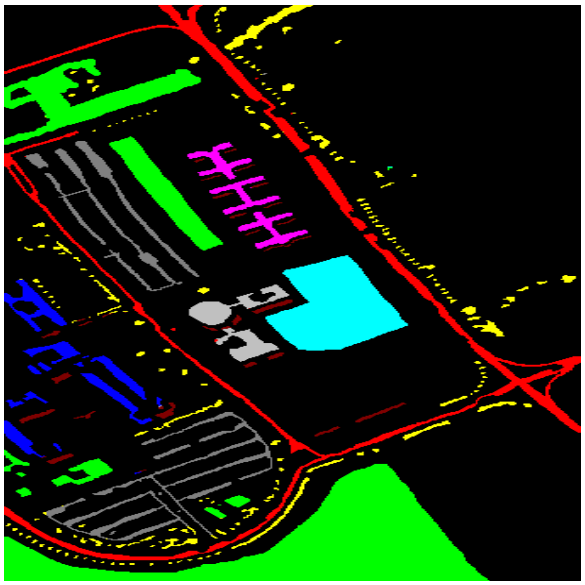


Classification Map

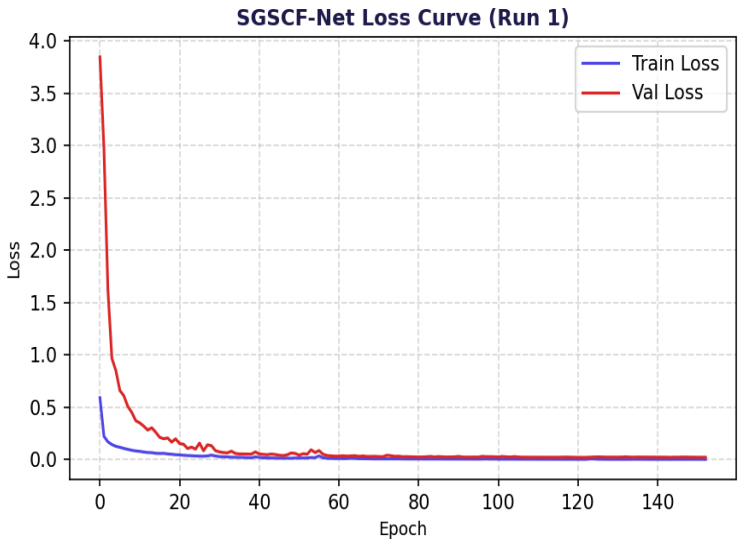


Loss Curve

Configuration: Patch 17x17 | Batch 256 | OA: 99.63%



Classification Map



Loss Curve